Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau
Module - II
Practical – VIII

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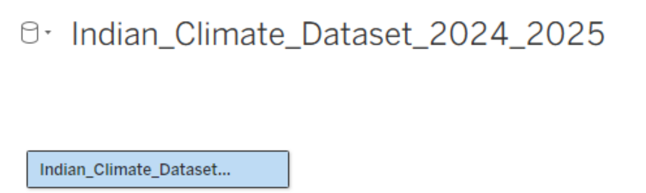
Class:TYIT	Batch:2
Date of Assignment: 27/08/26	Date/Time of Submission: 27/08/26

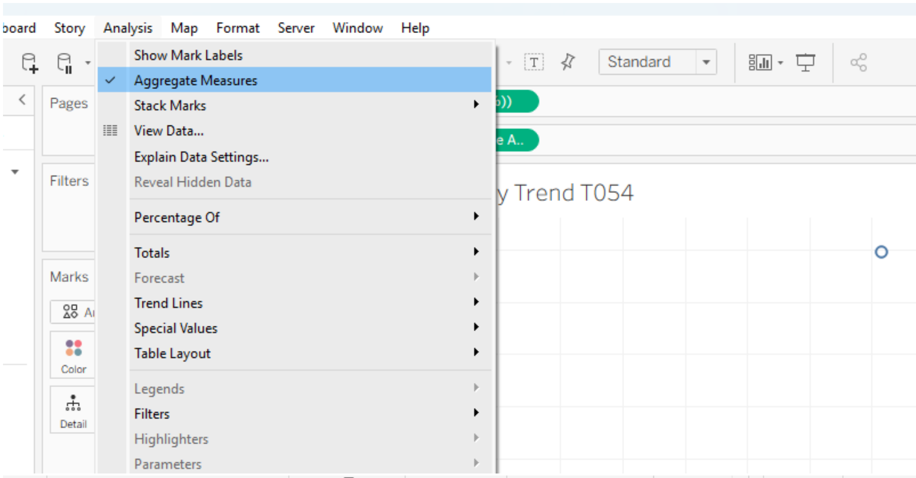
8. Performing Advanced Analytics using Trends, Clustering, and Forecasting.

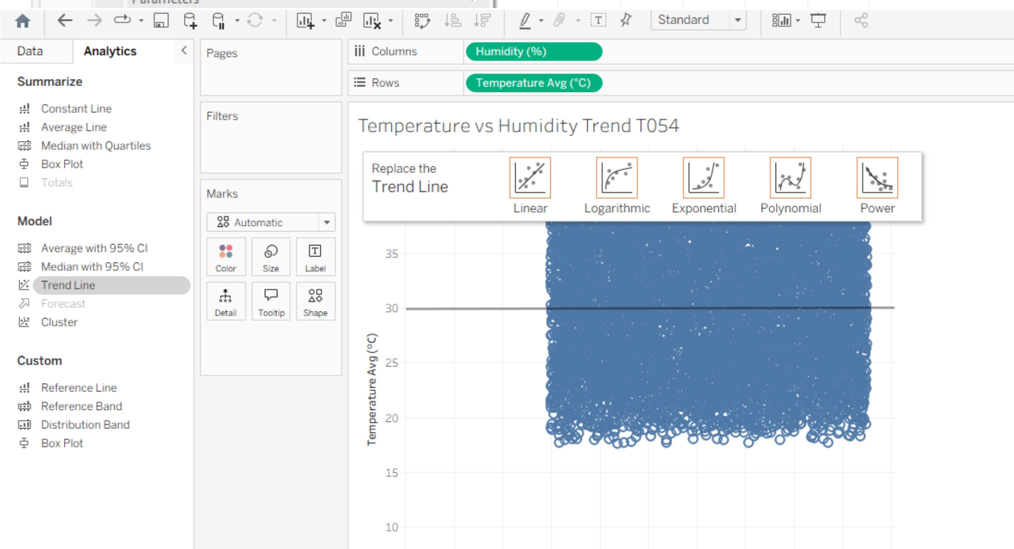
a) Add and interpret trend lines (linear, exponential).

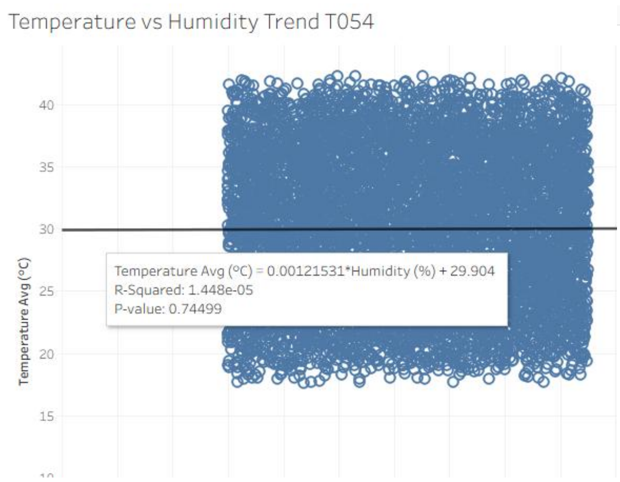
STEPS:

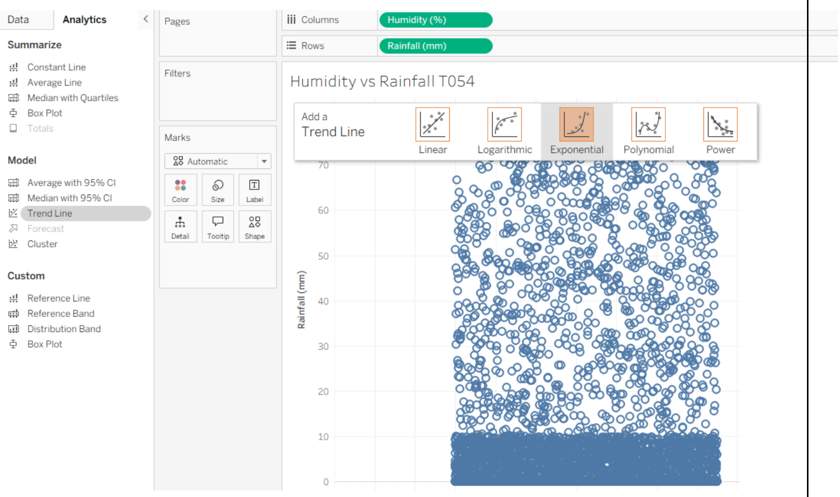
1. Load Indian_Climate_Dataset_2024_2025.csv.
2. Open a new worksheet and rename it to Temperature vs Humidity Trends.
3. Drag Humidity (%) to the Columns shelf and Drag Temperature_Avg (°C) to the Rows shelf.
5. go to Analysis and uncheck Aggregate Measures
6.Switch to the Analytics pane drag Trend Line onto the canvas and drop it onto Linear.
7.It displays the linear equation: Temperature_avg,R squared and P-value.
8. Open new sheet Drag Humidity (%) to Columns and Rainfall (mm) to Rows.
9. Switch to the Analytics pane drag Trend Line onto the canvas and drop it onto exponential.
10. It displays the Rainfall, R squared and P-value.

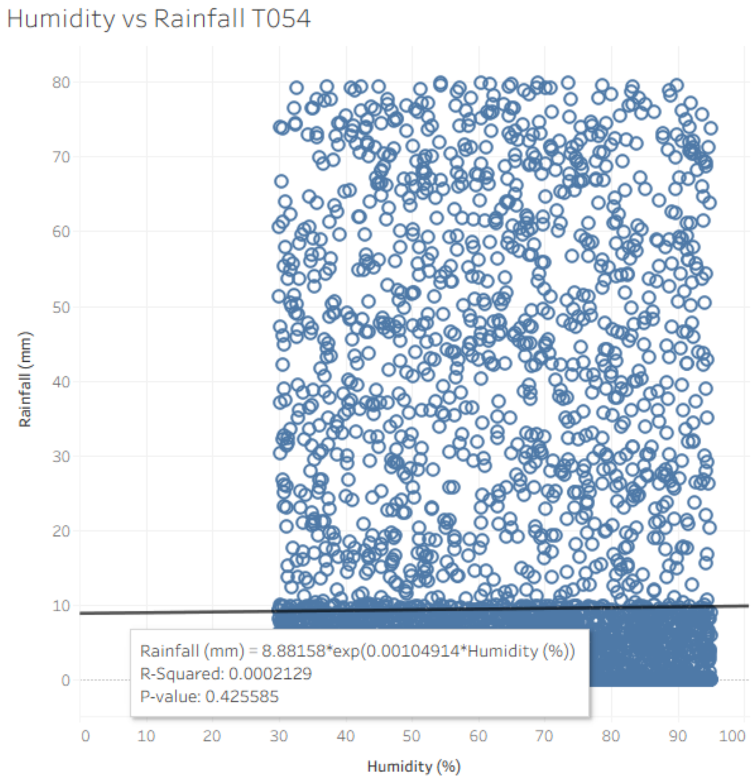












b) Customize and analyze trend models.

STEPS:-

1.Open your Temperature vs Humidity Trends sheet.

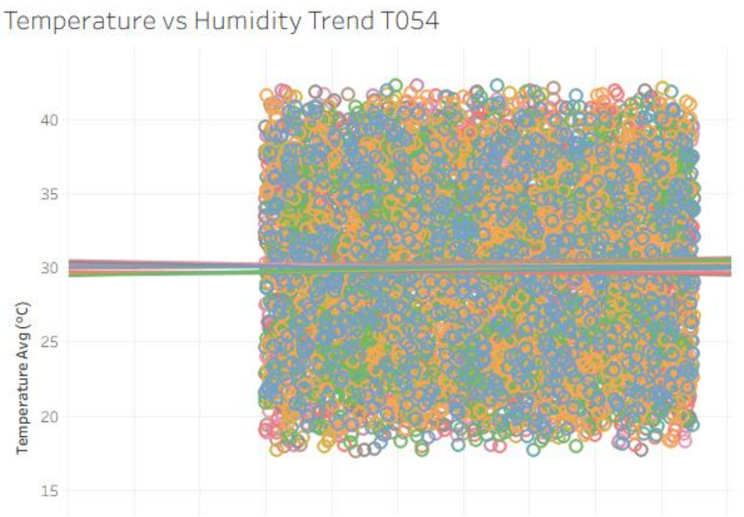
2. Drag City from the Data pane onto Color on the Marks card.

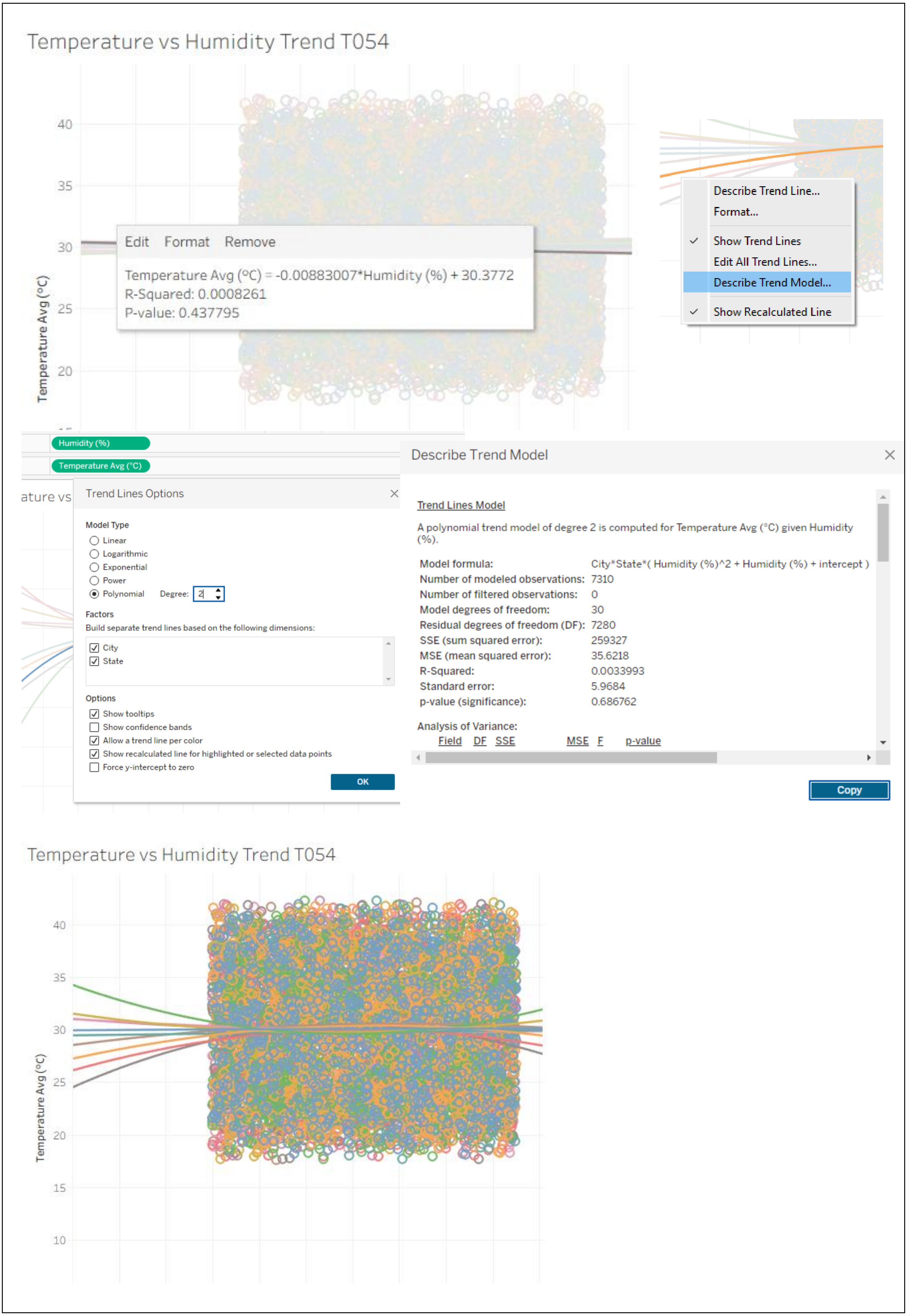
3. Right-click on any trend line in the view and select Edit Trend Lines.

4. In the options select Polynomial (Degree 2) and click OK.

5. Right-click anywhere on the chart background and select Describe Trend Line.

6. A detailed window will pop up showing the ANOVA table, Degrees of Freedom, Sum of Squared Errors (SSE), and F-statistic.




Temperature vs Humidity Trend T054
Temperature Avg (°C)
40
35
30
25
20
Edit Format Remove
Temperature Avg (°C) = -0.00883007*Humidity (%) + 30.3772
R-Squared: 0.0008261
P-value: 0.437795
Describe Trend Line...
Format...
Show Trend Lines
Edit All Trend Lines...
Describe Trend Model...
Show Recalculated Line
Humidity (%)
Temperature Avg (°C)
Trend Lines Options
Model Type
Linear
Logarithmic
Exponential
Power
Polynomial
Degree: 2
Factors
Build separate trend lines based on the following dimensions:
City
State
Options
Show tooltips
Show confidence bands
Allow a trend line per color
Show recalculated line for highlighted or selected data points
Force y-intercept to zero
OK
Describe Trend Model
Trend Lines Model
A polynomial trend model of degree 2 is computed for Temperature Avg (°C) given Humidity (%).
Model formula: City*State*(Humidity (%)^2 + Humidity (%) + intercept)
Number of modeled observations: 7310
Number of filtered observations: 0
Model degrees of freedom: 30
Residual degrees of freedom (DF): 7280
SSE (sum squared error): 259327
MSE (mean squared error): 35.6218
R-Squared: 0.0033993
Standard error: 5.9684
p-value (significance): 0.686762
Analysis of Variance:
Field DF SSE MSE F p-value
Copy
Temperature vs Humidity Trend T054
Temperature Avg (°C)
40
35
30
25
20
15
10

c) Perform clustering on datasets.

STEPS:

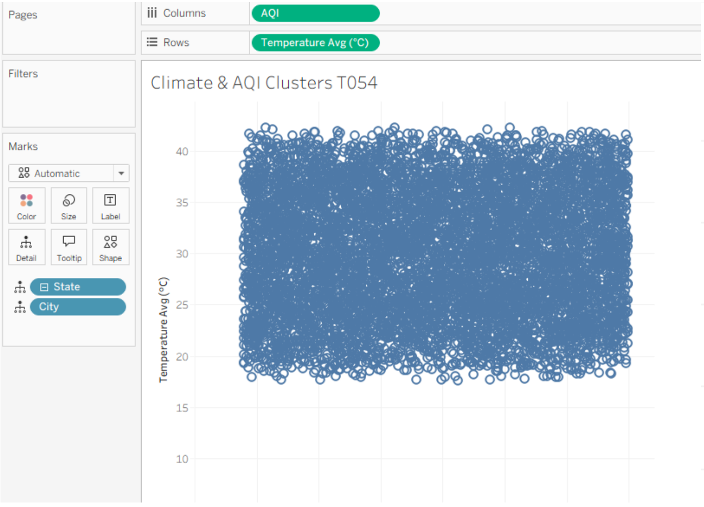
1. Open a new worksheet and name it Climate & AQI Clusters.
2. Drag AQI to Columns and Temperature_Avg (°C) to Rows.
3. go to Analysis and uncheck "Aggregate Measures" and Drag City to Detail on the Marks card.
4. Switch to the Analytics pane drag Cluster drag and drop the variables on the box):

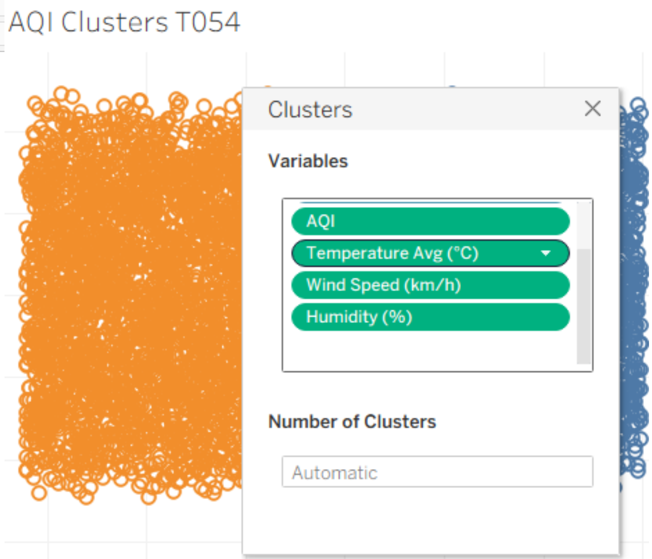
AQI , Temperature_Avg (°C), Humidity (%) , Wind_Speed (km/h) and Number of Clusters, leave it as Automatic.

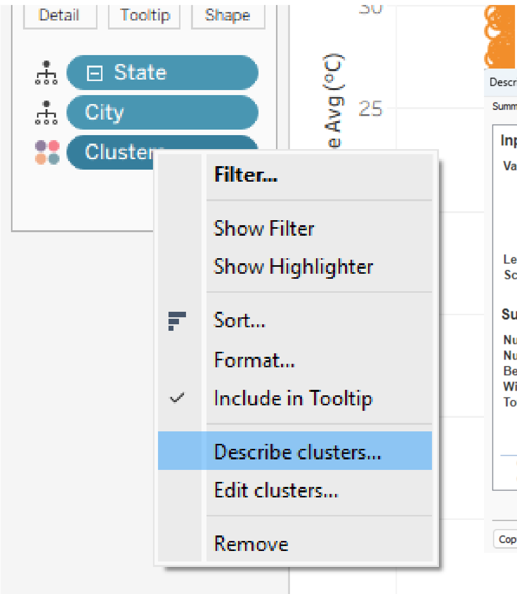
5. Right-click on the Clusters field the popup window, review two key tabs:

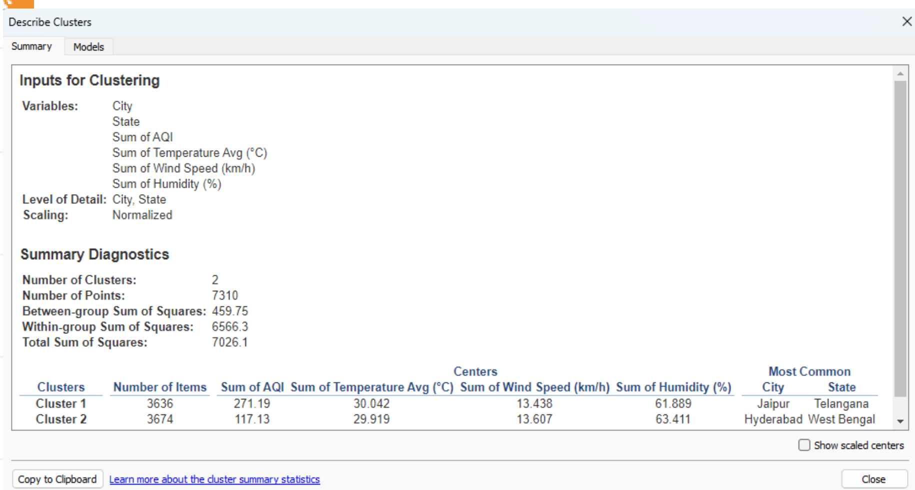
Summary: Shows the mean value of each metric (AQI, Temperature, Humidity, Wind Speed) for each cluster.

Models: Shows ANOVA statistics indicating which variables contributed most heavily to separating the clusters.









Describe Clusters

Summary | Models

Inputs for Clustering

Variables:	City State Sum of AQI Sum of Temperature Avg (°C) Sum of Wind Speed (km/h) Sum of Humidity (%)
Level of Detail:	City, State
Scaling:	Normalized

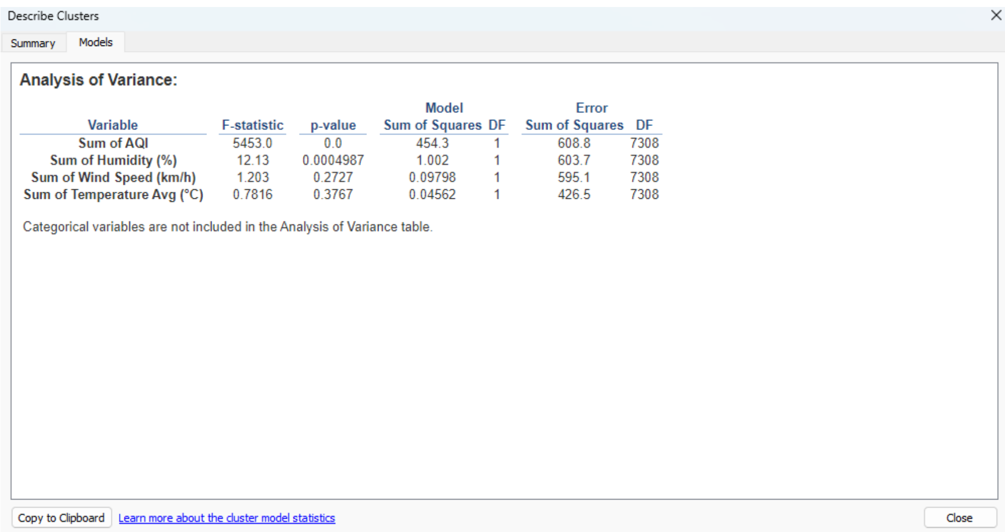
Summary Diagnostics

Number of Clusters:	2
Number of Points:	7310
Between-group Sum of Squares:	459.75
Within-group Sum of Squares:	6566.3
Total Sum of Squares:	7026.1

		Centers				Most Common	
Clusters	Number of Items	Sum of AQI	Sum of Temperature Avg (°C)	Sum of Wind Speed (km/h)	Sum of Humidity (%)	City	State
Cluster 1	3636	271.19	30.042	13.438	61.889	Jaipur	Telangana
Cluster 2	3674	117.13	29.919	13.607	63.411	Hyderabad	West Bengal

Show scaled centers

Copy to Clipboard | Learn more about the cluster summary statistics | Close



Describe Clusters

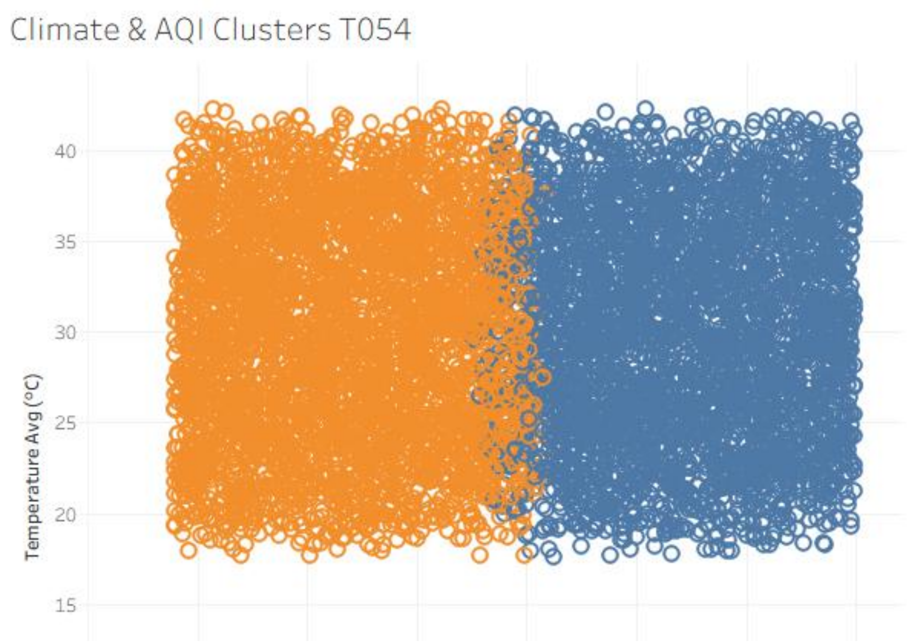
Summary | Models

Analysis of Variance:

			Model		Error	
Variable	F-statistic	p-value	Sum of Squares	DF	Sum of Squares	DF
Sum of AQI	5453.0	0.0	454.3	1	608.8	7308
Sum of Humidity (%)	12.13	0.0004987	1.002	1	603.7	7308
Sum of Wind Speed (km/h)	1.203	0.2727	0.09798	1	595.1	7308
Sum of Temperature Avg (°C)	0.7816	0.3767	0.04562	1	426.5	7308

Categorical variables are not included in the Analysis of Variance table.

Copy to Clipboard | Learn more about the cluster model statistics | Close



d) Analyze distributions using built-in tools.

STEPS:

1.Open a new worksheet and name it Temperature Distributions.

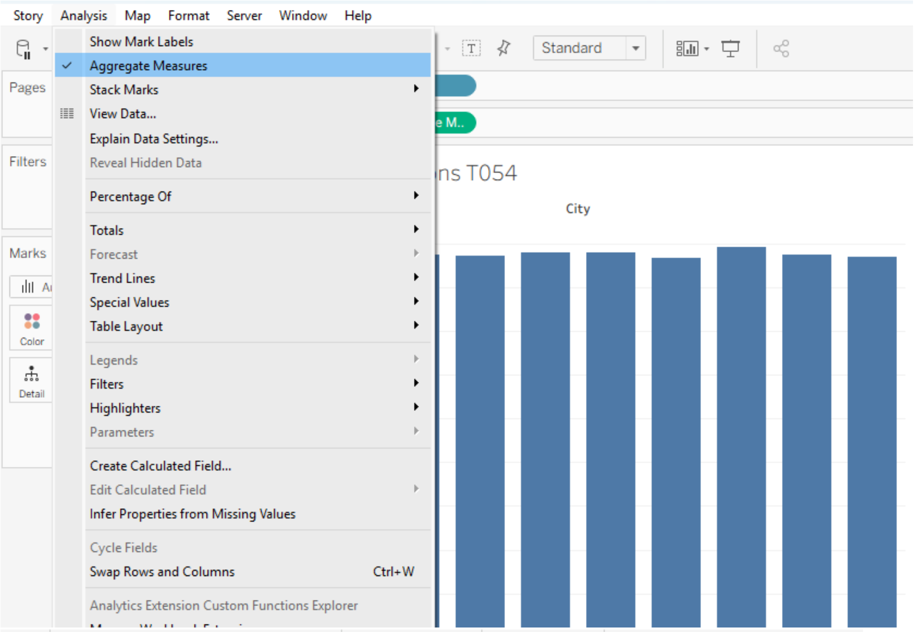
2. Drag City to Columns and Drag Temperature_Max (°C) to Rows.

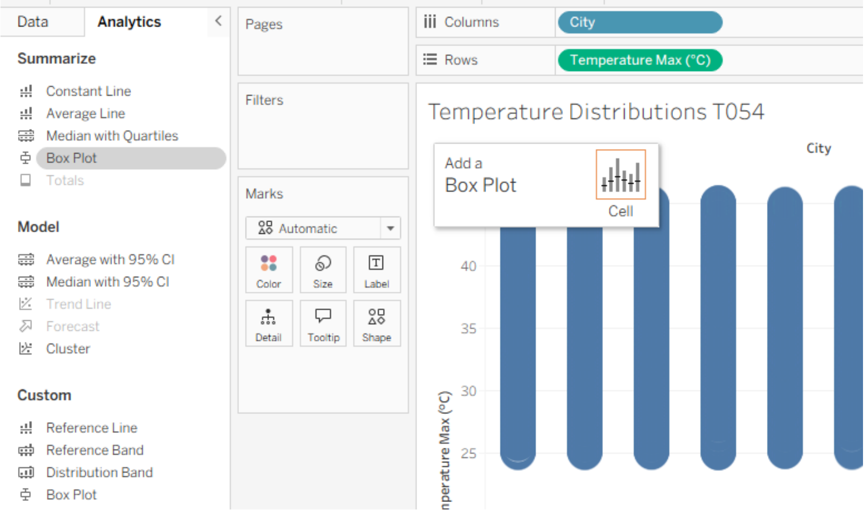
3. Switch to the Analytics pane drag Box Plot onto the canvas and drop it on the chart.

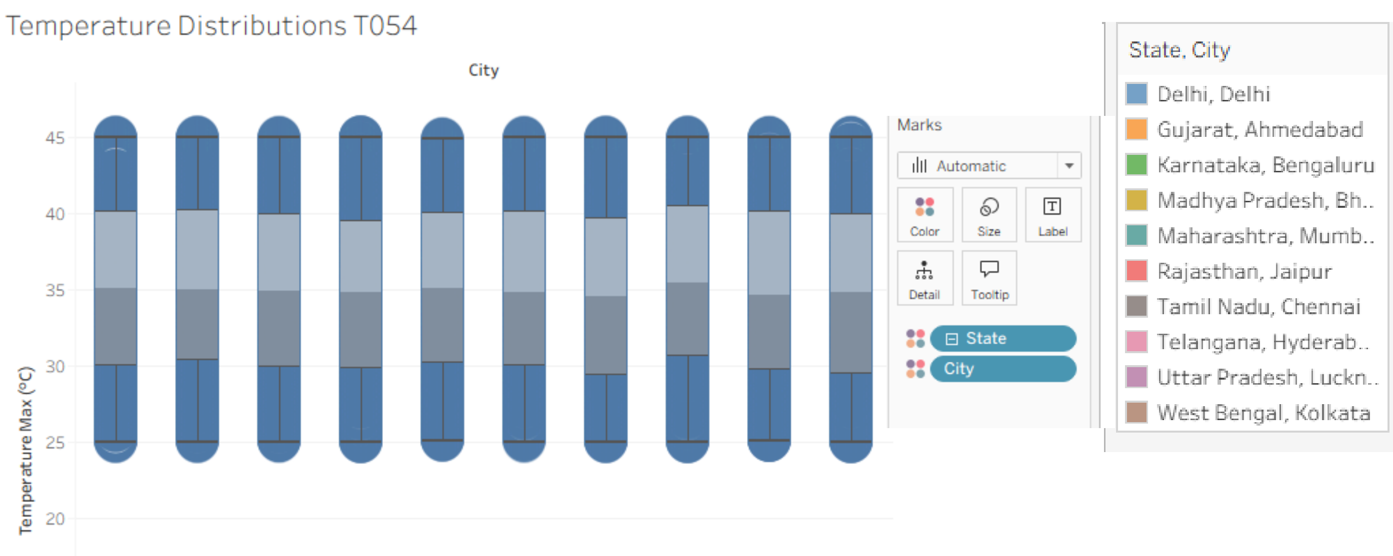
4. Open a new worksheet named AQI Histogram. From the Data Pane, click on AQI.

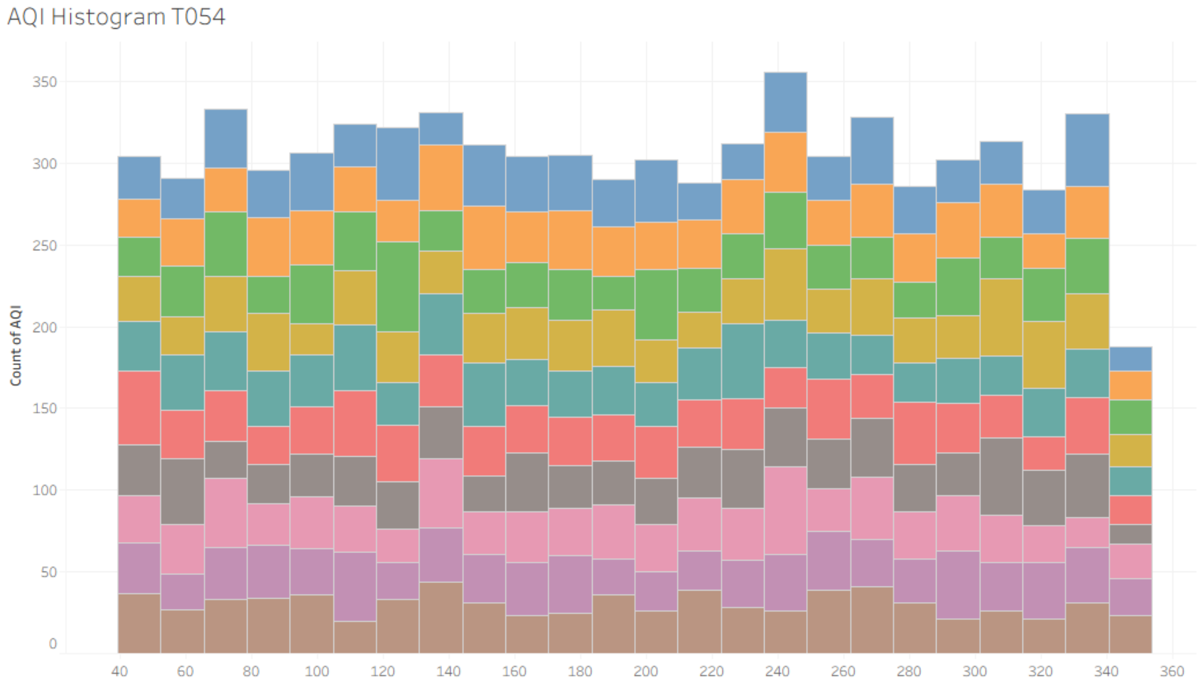
5. Open the Show Me panel in the top-right corner.Select the Histogram icon.

6. Drag City to Color on the Marks card





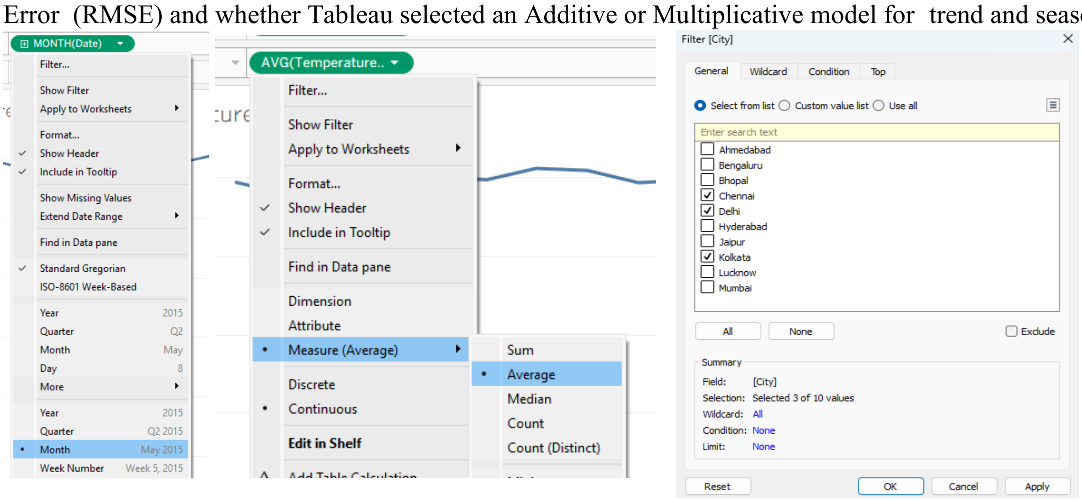


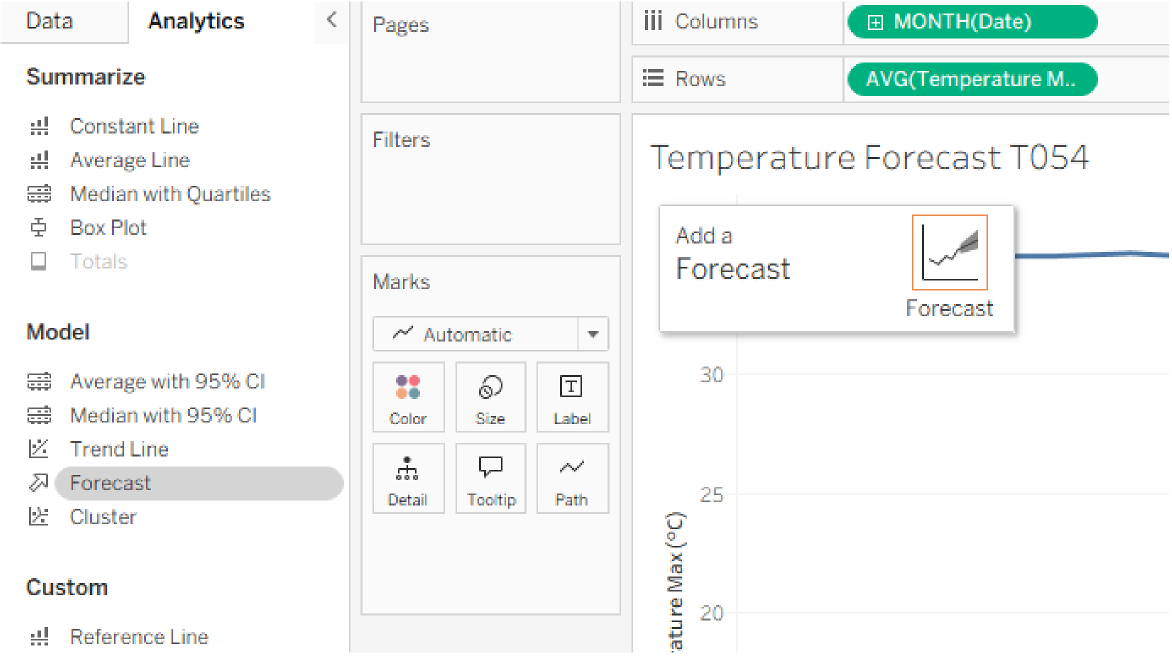


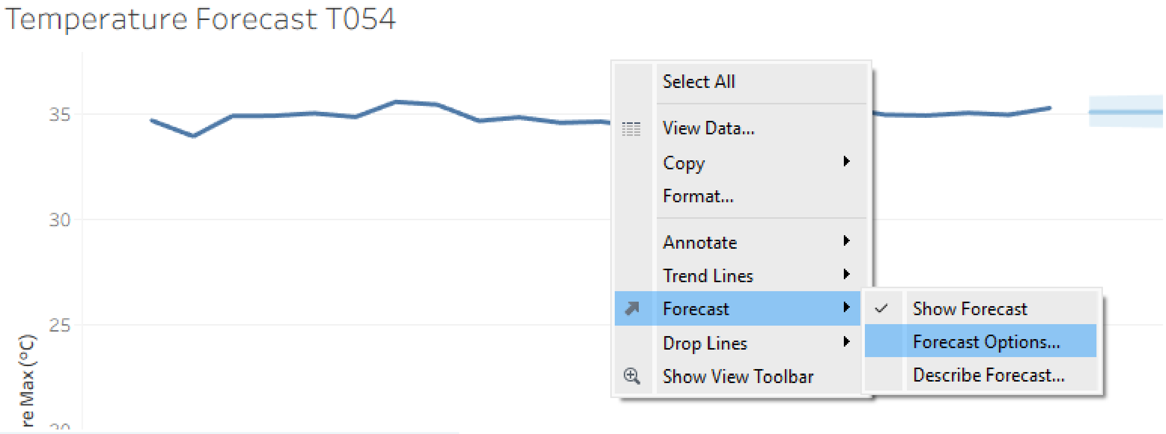
e) Apply forecasting techniques.

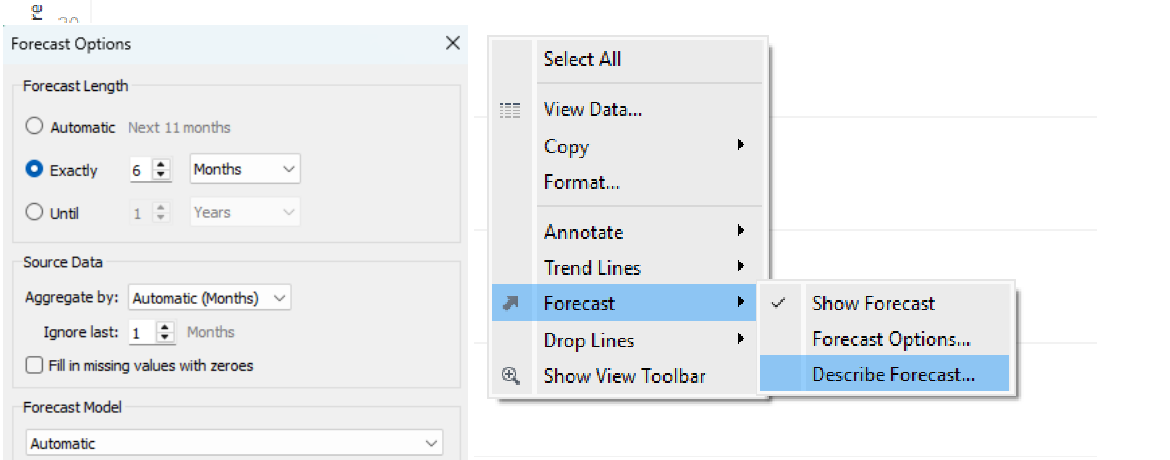
STEPS:

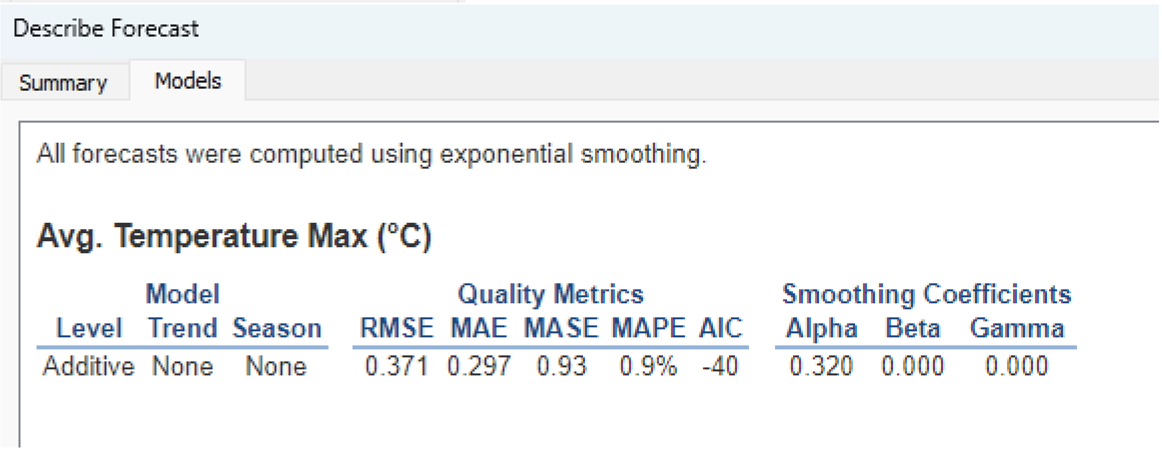
1. Open a new worksheet and name it Temperature Forecast.
2. Drag Date to Columns. change it to continuous Month.
3. Drag Temperature_Max (°C) to Rows and Drag City to Filters.
4. Switch to the Analytics Pane drag Forecast onto the canvas and drop it.
5.Right-click anywhere on the chart canvas => select Forecast =>Forecast Options.
6. set it to Exactly 6 Months into the future.
7. Right-click the chart => Forecast =>Describe Forecast and Review the Models tab to check metrics like the Root Mean Squared Error (RMSE) and whether Tableau selected an Additive or Multiplicative model for trend and seasonality.










Describe Forecast

Summary Models

All forecasts were computed using exponential smoothing.

Avg. Temperature Max (°C)

Model			Quality Metrics					Smoothing Coefficients		
Level	Trend	Season	RMSE	MAE	MASE	MAPE	AIC	Alpha	Beta	Gamma
Additive	None	None	0.371	0.297	0.93	0.9%	-40	0.320	0.000	0.000

f) Compare different statistical models.

STEPS:

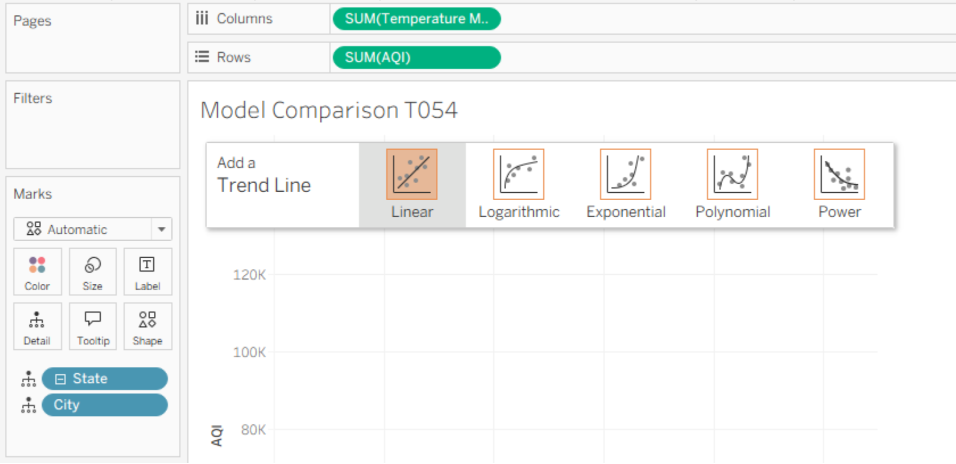
1.Open a new worksheet named Model Comparison.

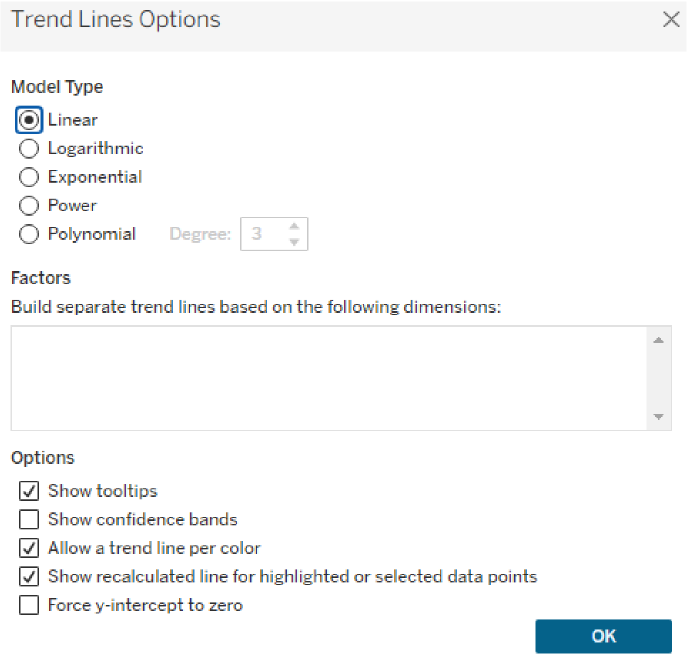
2. Drag Temperature_Max (°C) to Columns and Drag AQI to Rows.

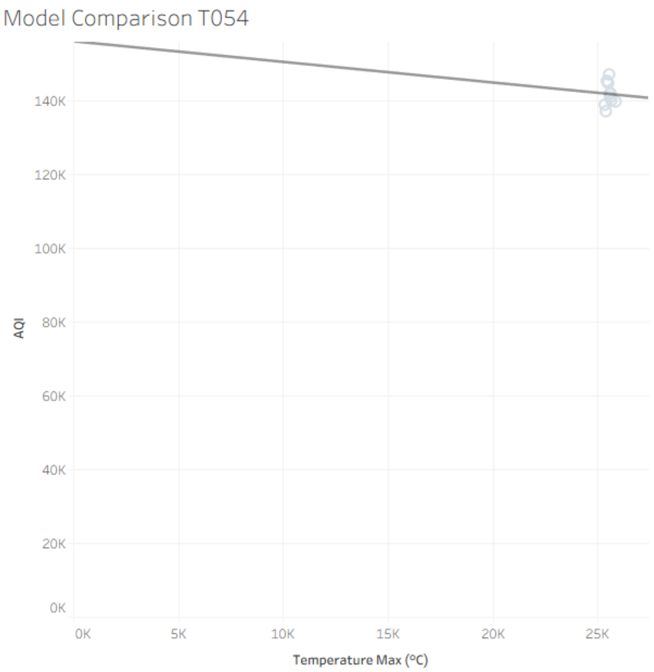
3. Drag City to Detail on the Marks card.

4. Open the Analytics Pane and Drag Trend Line onto the canvas and drop it on Linear.

5. Compare the R^2value for each model type. The model with the higher R^2 and a p-value < 0.05 provides the best fit.







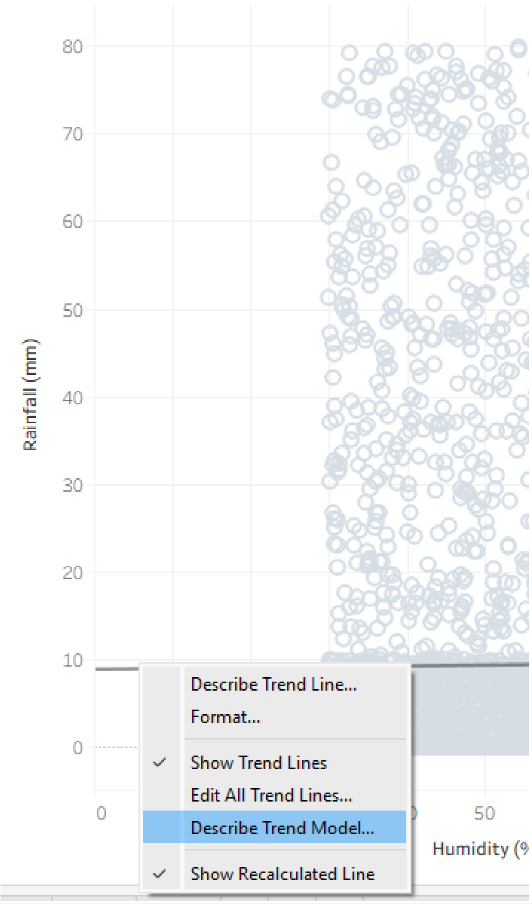
g) Export and interpret statistical results.

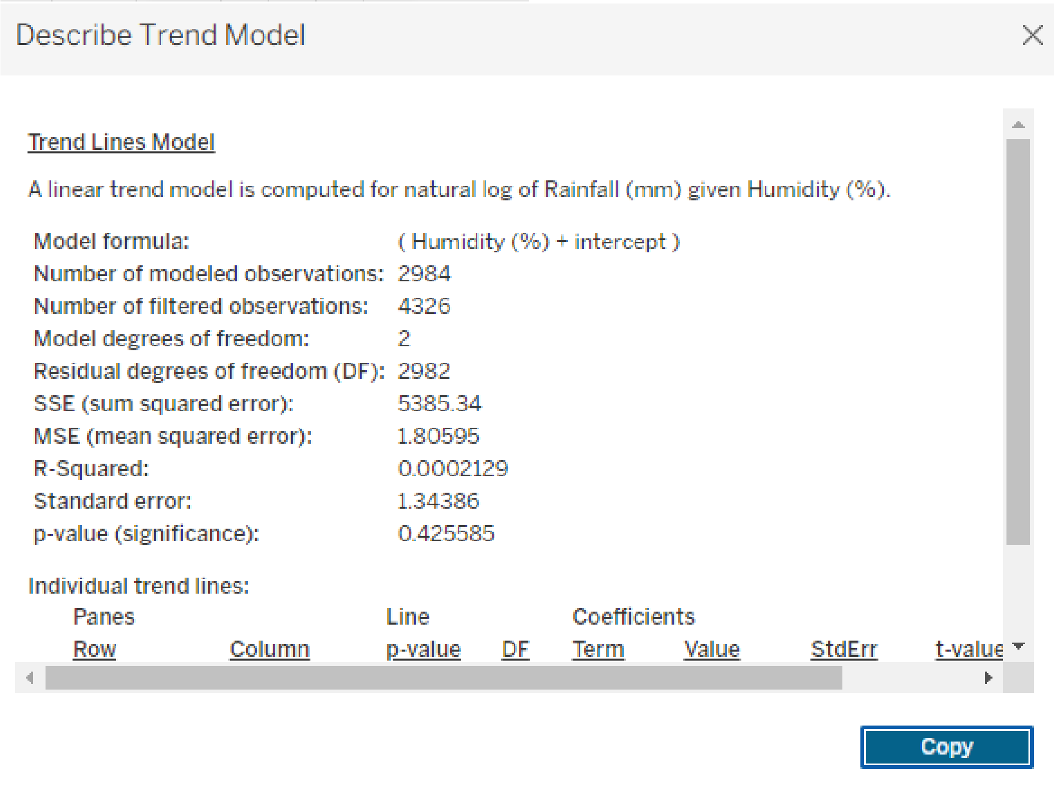
STEPS:

1.On Humidity vs Rainfall worksheet ,right-click the trend line or canvas.

2. Select Describe Trend Model. In the pop-up window, click Copy at the bottom.

3. now we have the full ANOVA summary table and parameter estimates copied to your clipboard





Describe Trend Model

Trend Lines Model

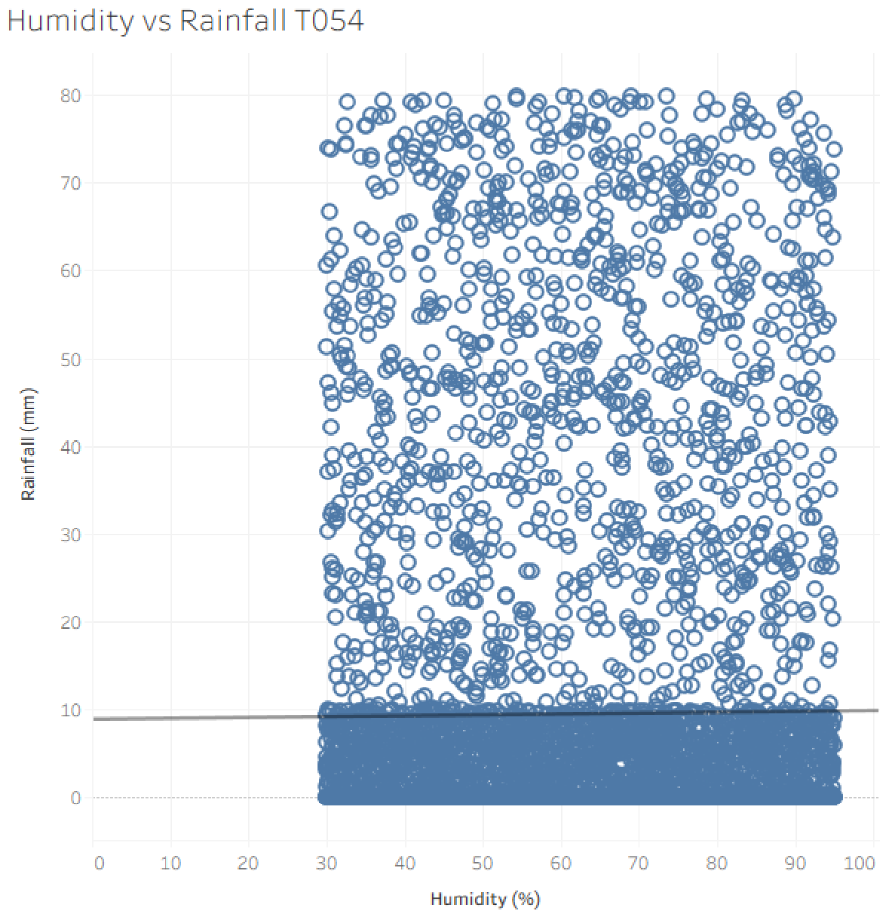
A linear trend model is computed for natural log of Rainfall (mm) given Humidity (%).

Model formula:	(Humidity (%) + intercept)
Number of modeled observations:	2984
Number of filtered observations:	4326
Model degrees of freedom:	2
Residual degrees of freedom (DF):	2982
SSE (sum squared error):	5385.34
MSE (mean squared error):	1.80595
R-Squared:	0.0002129
Standard error:	1.34386
p-value (significance):	0.425585

Individual trend lines:

Panes		Line		Coefficients			
Row	Column	p-value	DF	Term	Value	StdErr	t-value

Copy



CONCLUSION:

The advanced analytics features of Tableau were successfully applied to the dataset. Trend lines, clustering, distribution analysis, and forecasting were used to identify patterns and relationships in the data. Different statistical models were compared, and the analytical results were exported and interpreted to gain meaningful insights for better data-driven decision-making.